

Informal learning in everyday human–LLM interaction

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Abstract

As LLM-based assistants become everyday companions for work and study, life-long learning is increasingly mediated by everyday human–LLM interaction. Yet it remains unclear whether these interactions elicit the engagement through which people build understanding. We analyse 71,412 naturalistic WildChat conversations, adapting learning-science constructs to 577,225 turns through human-verified LLM-assisted annotation. Across 287,778 user turns, cognitive engagement appeared in 30.9% and constructive engagement—the stricter level in which users elaborate, test, revise or extend ideas—in 5.2%. Constructive engagement was shaped by learning framing, task ecology and sustained exchange. On the other side, assistant support was most informative not as a binary scaffolding signal but as a situated match between support form and user activity: feedback-like and explanatory forms aligned with constructive engagement, and adjacent-turn analyses showed local, state-dependent coupling rather than automatic accumulation. These findings make informal learning in everyday AI use measurable, laying a foundation for personal assistants that help people build knowledge while solving real problems.

Keywords: human–LLM interaction, informal learning, large language models, scaffolding, learner engagement

1 Introduction

Large language models (LLMs) are becoming embedded in the cognitive work of everyday life. People use them to write, code, search, troubleshoot and make sense of unfamiliar problems. Most of these interactions do not take place in classrooms, training programmes or purpose-built tutoring systems. Instead, they arise in the course of routine activity, when users turn to LLMs to solve an immediate problem, clarify a concept, improve a draft or navigate an unfamiliar task. In such moments, the model may do more than help complete the task at hand: it may shape what the user notices, understands and tries next. Everyday human–LLM interaction therefore constitutes a rapidly expanding setting for informal learning—that is, learning that occurs outside formal curricula and designed instructional environments, and instead emerges through everyday activity, self-directed inquiry and problem solving [1]. This learning may be intentional, when users explicitly seek understanding, or incidental, when understanding develops while pursuing another immediate goal [2, 3].

Yet existing research on LLMs and learning has focused primarily on settings in which learning support is deliberately designed, such as automated feedback, question generation, retrieval practice and tutoring systems [4–7]. These studies show that LLMs can enact instructional strategies when they are explicitly prompted, trained or embedded in purpose-built systems [8]; they tell us less about whether learning-oriented engagement emerges when general-purpose LLMs are used in the course of everyday tasks.

This gap matters because the educational significance of everyday AI use depends not only on what models can provide, but on what users do with that support. The key question is not simply whether LLMs can generate explanations, examples or feedback, but whether these interactions keep users active in the work of understanding: asking questions, testing assumptions, revising ideas and extending solutions. Answering this question would open a path toward future personal AI assistants that function not only as tools for completing tasks, but also as context-sensitive companions for lifelong learning. Such systems would need to recognize when a user is ready for feedback, explanation, modelling or direct help, and when support might instead short-circuit the user’s own sense-making. Conversely, without evidence about these interactional dynamics, it is difficult to know whether AI-mediated work is helping people build knowledge and skill over time, or merely making answer delivery faster while leaving human understanding opaque. We therefore ask how often users show learning-oriented engagement in everyday human–LLM interaction, and which contextual and interactional conditions make it more likely.

Studying this question in naturalistic logs requires adapting educational theory to data without fixed instructional tasks, explicit learning objectives or post-test outcomes. We translate theories of engagement and scaffolding into role-specific, turn-level behavioural signatures: user turns are coded for passive, active or constructive uptake [9, 10], while assistant turns are coded for the presence of scaffolded support and further characterized by support intent and support form [11–14]. Following the literature that deeper forms of overt engagement are more strongly linked to learning [10], we treat constructive engagement as the strictest observable indicator of learning-oriented participation available in naturalistic logs. These measures do not establish

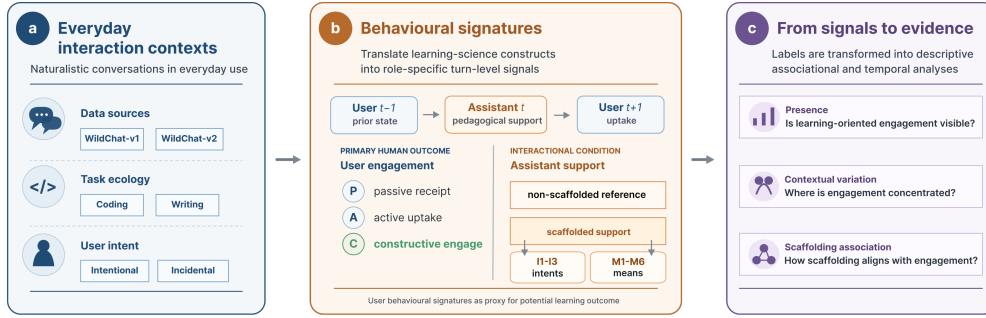


Fig. 1 Analytic framework for studying informal learning in everyday human-LLM interaction. **a**, Naturalistic WildChat conversations are situated by task ecology and user framing. **b**, Learning-science constructs are translated into turn-level behavioural signatures for user engagement and assistant support. **c**, The resulting labels support analyses of engagement presence, contextual variation and scaffolding-engagement association. Labels capture observable interactional signatures, not durable learning outcomes or causal instructional effects.

causal learning effects or durable knowledge gains; instead, they allow us to observe whether learning-oriented engagement is present in everyday use and to test which contextual and interactional conditions are associated with deeper user participation.

Using this adapted framework, we analyse 71,412 naturalistic WildChat conversations, focusing on coding and writing as two common task ecologies of everyday LLM use. The corpus contains 287,778 user turns and 289,447 assistant turns. The analysis proceeds in three layers: first, whether learning-oriented engagement is visible and how it is composed; second, where constructive engagement is concentrated across user framing, task ecology and interaction depth; and third, how assistant scaffolding is associated with constructive participation across conversation-level, support-form and adjacent-turn temporal scales.

We find, first, that learning-oriented engagement is a recurrent feature of everyday human-LLM interaction, although its constructive form is selective. Across WildChat, cognitive engagement appears in 30.9% of user turns and constructive engagement in 5.2%, while emotional engagement is rare; scaffolded support appears in 35.4% of assistant turns. These rates show that everyday AI use often involves more than answer delivery, but the most generative and constructive form of user engagement is a stricter and less frequent behavioural signal. Second, constructive engagement is organized by context: it is amplified by explicit learning framing, varies across task ecologies and becomes most visible in sustained exchanges where users return to, interrogate and reshape the assistant’s response. Third, assistant scaffolding marks richer constructive participation, but not as a uniform or treatment-like effect. Its association with engagement depends on user framing, support form, task-specific support supply and temporal scale: feedback-like and explanatory support align most clearly with constructive engagement, whereas more directive or questioning forms are less aligned with visible elaboration, and the strongest temporal evidence is local and state-dependent rather than cumulative. Together, these findings locate everyday human-LLM interaction in a middle space between ordinary tool use and designed instruction. By making

this middle space observable, the study provides a foundation for future personal AI assistants that support lifelong learning in the flow of everyday work—not by turning every interaction into a formal lesson, but by helping everyday human–AI collaboration become a site where people build, test and extend understanding while solving real problems.

2 Results

The results proceed from engagement presence to the conditions that differentiate its depth. We first ask whether everyday human–LLM conversations contain learning-oriented engagement, distinguishing broad cognitive uptake from the stricter constructive form. We then examine where constructive engagement is concentrated across user framing, task ecology and sustained exchange. Finally, we ask how assistant scaffolding relates to constructive engagement: first as a conversation-level condition, then through support forms, and finally through adjacent-turn temporal ordering. Because the logs contain no direct assessment of knowledge gain, engagement labels are interpreted as behavioural signatures of potential learning, and scaffolding analyses are interpreted as associations rather than causal evidence of learning effects.

2.1 Everyday human–LLM conversations contain learning-oriented engagement

Across the two WildChat task settings, 30.9% of user turns showed cognitive engagement, 5.2% reached constructive engagement, and emotional engagement was rare, appearing in only 0.73% of user turns (Table 1). This establishes the first descriptive pattern: everyday human–LLM interaction contains observable learning-oriented engagement, but that engagement is primarily cognitive rather than affective, and only a smaller subset reaches the more generative constructive form. In other words, users often did more than simply request or receive task output, but the depth of their engagement varied.

Figure 2a decomposes cognitively engaged turns into passive receipt, active uptake and constructive engagement. Active uptake accounted for 79.5% to 80.4% of cognitively engaged turns across the two WildChat task settings, constructive engagement for 13.6% to 18.4%, and passive receipt for 1.2% to 6.9%. Thus, learning-oriented engagement was not a binary signal but a depth-graded pattern. Most cognitively engaged turns involved active uptake: users used, applied or followed up on the assistant’s response. A smaller but stable subset involved constructive engagement, in which users elaborated, tested, revised or extended ideas. This distinction matters for the rest of the Results. Cognitive engagement provides the broad signal that everyday LLM use can become learning-oriented, active uptake shows that users often work with the assistant’s response, and constructive engagement identifies the stricter cases of deeper user participation that we seek to explain. The following analyses therefore focus on where this constructive form appears and which interactional conditions are associated with it.

Table 1 Dataset and behavioural overview. Summary statistics for the two WildChat task settings analysed in the study. Constructive engagement is reported as the stricter depth level within cognitive engagement and serves as the focal user-side outcome in subsequent analyses.

Setting	Sample size			User framing	Assistant support	Cognitive engagement		Emotional engagement
	Conversations	User turns	Assistant turns	Intentional conv. (%)	Scaffolded turns (%)	Overall (%)	Constructive level (%)	Emotional (%)
WildChat coding	31,878	124,073	124,623	37.13	51.03	50.21	9.23	0.95
WildChat writing	39,534	163,705	164,824	25.80	23.53	16.34	2.22	0.57
Total / pooled	71,412	287,778	289,447	30.86	35.37	30.9	5.2	0.73

Note. Percentages for cognitive, constructive and emotional engagement are computed over user turns; percentages for scaffolded support are computed over assistant turns. The constructive-level column reports the share of all user turns that reached the constructive depth level within cognitive engagement. Figure 2a reports the composition of passive, active and constructive uptake among cognitively engaged turns.

2.2 Constructive engagement is shaped by user framing, task ecology and interaction depth

Having established that everyday human–LLM conversations contain learning-oriented engagement, we next examined where its constructive form was concentrated. User framing provided the first contrast (Fig. 2b). Across the two WildChat task settings, conversations explicitly oriented toward learning or exploration showed higher engagement than conversations without such framing. Cognitive engagement increased from 10.5–32.5% in unintentional conversations to 28.2–76.6% in intentional conversations, and constructive engagement increased from 1.5–5.7% to 5.2–13.1%. Explicit learning or exploratory framing therefore amplified constructive engagement, but did not create it from zero: even task-oriented conversations without explicit learning intent contained constructive participation.

Task ecology provided a second source of variation. In the two everyday task ecologies sampled here, coding-oriented conversations showed higher engagement than writing-oriented conversations. Cognitive engagement was 50.21% in coding conversations, compared with 16.34% in writing conversations; constructive engagement was 9.23% in coding and 2.22% in writing. Importantly, this contrast was not reducible to explicit learning framing: unintentional coding conversations showed higher cognitive engagement than intentional writing conversations. This pattern suggests that task ecology was not merely a backdrop for explicit learning intent; it shaped the baseline opportunities for users to ask questions, test assumptions, refine solutions and extend ideas. One plausible reason is that coding tasks often make uncertainty and failure externally visible through errors, constraints and executable tests, whereas writing tasks may more often be resolved through drafting, editing or applying a proposed revision without requiring users to articulate the underlying rationale [15–17].

Interaction depth provided a third lens on where constructive participation surfaced (Fig. 2c). Across the two WildChat task settings, the probability that a conversation contained at least one constructive user turn increased monotonically with behavioural depth, from 0.060–0.151 in 2–3-turn conversations to 0.150–0.463 in conversations with seven or more turns. The gradient was steeper in coding-oriented conversations, but the same direction appeared in writing-oriented conversations as well. Because this measure captures whether a constructive turn occurred at least once

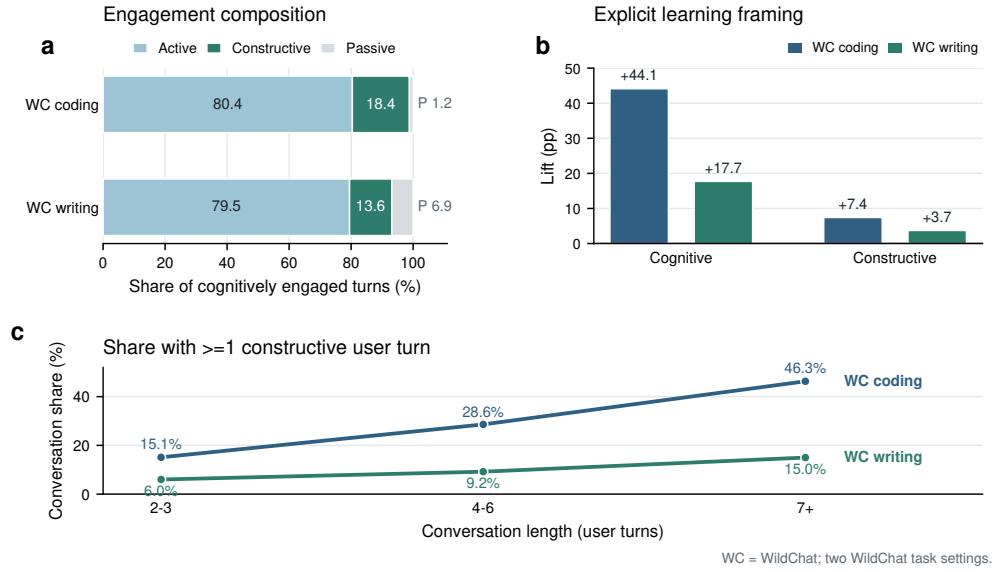


Fig. 2 Learning-oriented engagement varies by engagement form, user framing and conversation length. **a**, Composition of cognitively engaged user turns by active, constructive and passive levels in the two WildChat task settings. **b**, Cognitive and constructive engagement rates in conversations with versus without explicit learning framing. **c**, Share of conversations with at least one constructive user turn by conversation-length bucket. Panel a uses cognitively engaged turns as the denominator; panel b reports percentages of user turns; panel c reports conversation-level shares. WC, WildChat.

in a conversation, longer exchanges naturally provide more opportunities for such turns to appear and may also reflect more complex tasks or more engaged users. We therefore do not interpret depth as evidence that turns accumulate into learning progress by themselves. The substantive point is instead that constructive engagement is most visible when everyday LLM use moves beyond one-off answer delivery and becomes a sustained exchange in which users return to, interrogate and reshape the assistant’s response.

Together, these results show that constructive engagement is ecological. It is amplified by explicit learning framing, varies across task ecologies and concentrates in sustained exchanges. This motivates the next step of the analysis: asking whether assistant scaffolding is one interactional condition associated with richer constructive participation.

2.3 Scaffolded conversations show richer constructive participation

We next asked whether assistant scaffolding was associated with richer constructive participation. At the conversation level, conversations containing at least one scaffolded assistant turn had higher constructive ratios than conversations without scaffolded support in both WildChat task settings (Fig. 3a). The raw differences were

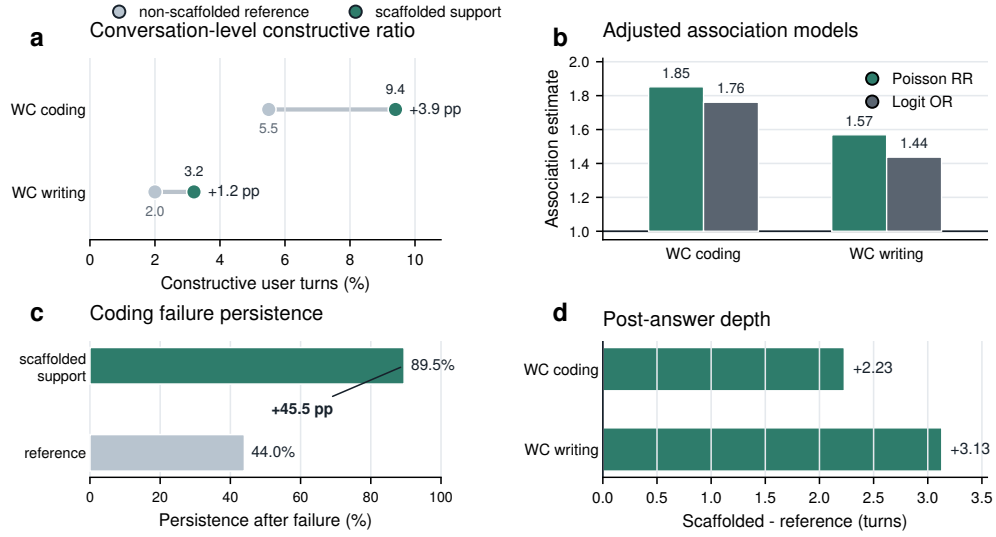
+3.9 percentage points in coding-oriented conversations and +1.2 points in writing-oriented conversations. Scaffolded conversations also showed greater post-answer depth in both task settings, with differences of +2.23 and +3.13 turns after the first assistant response, respectively (Fig. 3d). These patterns show that scaffolding co-occurred with both more constructive participation and more sustained exchange. However, because scaffolded support was observed rather than assigned, these raw contrasts could also reflect the contexts in which scaffolding appeared, including task complexity, user framing and interactional opportunity.

We therefore used covariate-adjusted models as a partial check on this compositional ambiguity, asking whether scaffolded-support presence remained associated with constructive participation after conditioning on observed contextual and interactional features. The association remained positive in both WildChat task settings. Poisson rate ratios for constructive-turn counts were 1.852 in coding and 1.569 in writing, and logistic odds ratios for whether a conversation contained at least one constructive turn were 1.761 and 1.437, respectively (Fig. 3b). These estimates indicate that scaffolded support was not fully explained by measured differences in conversation composition, while still remaining associational rather than causal.

Consistent with the context results above, the support–engagement association also varied by user framing. Stratified models indicated that this moderation was task-specific rather than uniform: the association between scaffolded support and constructive participation differed by intentional versus unintentional framing, but not in the same way across coding- and writing-oriented conversations. Scaffolding therefore did not operate as an isolated property of the assistant response. Its association with constructive engagement depended on the broader interactional state of the conversation.

We also used persistence after failure as a complementary learning-relevant signal of sustained engagement. In everyday informal learning, difficulty is a diagnostic moment because the user can either disengage, ask for another answer or continue working through the problem. Persistence after failure therefore complements constructive engagement by capturing whether the user remains in a repair-and-sense-making loop when the initial attempt does not work. This measure is most interpretable in coding, where breakdowns are more directly observable through errors, failed execution, debugging language and error-resolution contexts. In WildChat coding, scaffolded conversations showed substantially higher persistence after failure than conversations without scaffolded support: 0.895 versus 0.440 (Fig. 3c). Exploratory analyses in writing-oriented conversations showed similar directional patterns, but are reported in the Appendix because difficulty markers in writing are less consistently identifiable. Thus, scaffolded support was associated not only with richer constructive participation overall, but also with greater persistence after observable breakdowns in coding conversations.

Taken together, these results position scaffolded support as a meaningful interactional condition for constructive participation, but not as a stand-alone explanation for engagement. Scaffolded conversations were more constructive and deeper, adjusted associations remained positive, and coding-oriented conversations showed greater persistence after observable breakdowns. Yet these patterns varied by task ecology and



Panel c reports persistence after failure in coding-oriented WildChat conversations.

Fig. 3 Scaffolded WildChat conversations are associated with richer constructive participation. **a**, Constructive user-turn ratios in conversations with versus without scaffolded support. **b**, Covariate-adjusted associations between scaffolded-support presence and constructive participation. **c**, Persistence after failure in coding-oriented conversations, where breakdowns are more directly observable. **d**, Difference in post-answer interaction depth after the first assistant response. Numbers indicate scaffolded-minus-reference differences, except panel b, which reports Poisson rate ratios and logistic odds ratios.

Table 2 Summary of key support–engagement associations in WildChat.

Constructive-ratio contrasts compare conversations with versus without scaffolded support. Adjusted models report the association of scaffolded-support presence with constructive-turn count (Poisson rate ratio) and with having at least one constructive turn (logit odds ratio). Adjacent-turn lift is the difference in the probability that the next user turn is constructive after scaffolded versus non-scaffolded reference assistant turns.

Setting	Constructive ratio Has S2 / No S2	Difference (pp)	Depth diff. (turns)	Poisson RR	Logit OR	Adjacent lift (pp)	Around first S2 diff.
WildChat coding	0.094 / 0.055	+3.9	+2.23	1.852	1.761	+2.68	+0.014
WildChat writing	0.032 / 0.020	+1.2	+3.13	1.569	1.437	+1.13	-0.042

user framing. Scaffolding therefore appears most informative when considered as part of a broader interactional ecology: what users are trying to do, what the task makes visible and how support is delivered all shape whether users continue working constructively with ideas. This heterogeneity motivates a finer-grained analysis of scaffolding itself: whether different forms of support carry different signals for constructive engagement.

2.4 Support forms distinguish broad cognitive uptake from constructive engagement

The conversation-level association did not imply that all forms of scaffolding carried the same engagement signal. We therefore decomposed scaffolded support into the six support forms in the annotation scheme (M1–M6) and compared scaffolded conversations containing each form with scaffolded conversations that did not contain that form, stratified by intentional versus unintentional learning orientation. Because support-form labels were not mutually exclusive, these contrasts are descriptive associations rather than effects of isolated instructional moves. Even with this constraint, the pattern differentiated broad cognitive uptake from constructive engagement (Fig. 4a).

The overall cognitive-engagement panel shows how support forms drew users into learning-oriented participation. Feedback and hinting were more strongly associated with cognitive engagement in unintentional conversations than in intentional conversations (M1: +15.3 versus -3.0 percentage points; M2: +14.4 versus +6.3 points). This pattern suggests that evaluative feedback and partial cues may be especially useful for pulling otherwise task-oriented exchanges into cognitive work. Explanation, by contrast, showed the largest cognitive-engagement contrast in intentionally framed conversations (M4: +32.7 versus +16.8 points), consistent with this form aligning with users who are already seeking understanding. Modelling was positive in both framing groups and showed little framing difference (M5: +8.8 versus +9.8 points). Instructing and questioning showed a different profile: both were negatively associated with cognitive engagement in intentional and unintentional conversations, with larger negative contrasts in intentionally framed conversations. A plausible interactional explanation is that detailed instructions can make the next step executable without requiring users to articulate or revise their understanding, while questions in everyday assistant use may often function as clarification requests rather than designed Socratic prompts.

The constructive-engagement panel sharpened this distinction by separating broad cognitive uptake from more generative participation. Feedback-like support and explanation showed the strongest constructive-engagement contrasts, especially in intentionally framed conversations (M1: +8.5 versus +5.3 percentage points; M4: +8.6 versus +4.1 points). This is notable because feedback did not show the same pattern for overall cognitive engagement in intentional conversations, suggesting that its contribution is not simply to increase participation in general. Rather, feedback may be especially aligned with constructive engagement because it responds to a user's prior attempt and creates an occasion for revision, evaluation or extension. Explanation was similarly aligned with constructive engagement, consistent with the idea that making mechanisms visible can support users in elaborating or testing their understanding. Hinting was positive but smaller (M2: +4.4 versus +2.5 points), whereas modelling was positive but weak (M5: +0.9 versus +1.3 points). By contrast, instructing and questioning were associated with lower constructive engagement in both framing groups, with stronger negative contrasts in intentionally framed conversations (M3: -3.7 versus -0.8 points; M6: -4.8 versus -1.7 points). Thus, the support forms most aligned with constructive engagement were those that preserved room for users to evaluate, revise and extend ideas, rather than those that primarily prescribed the next step or shifted the exchange into clarification.

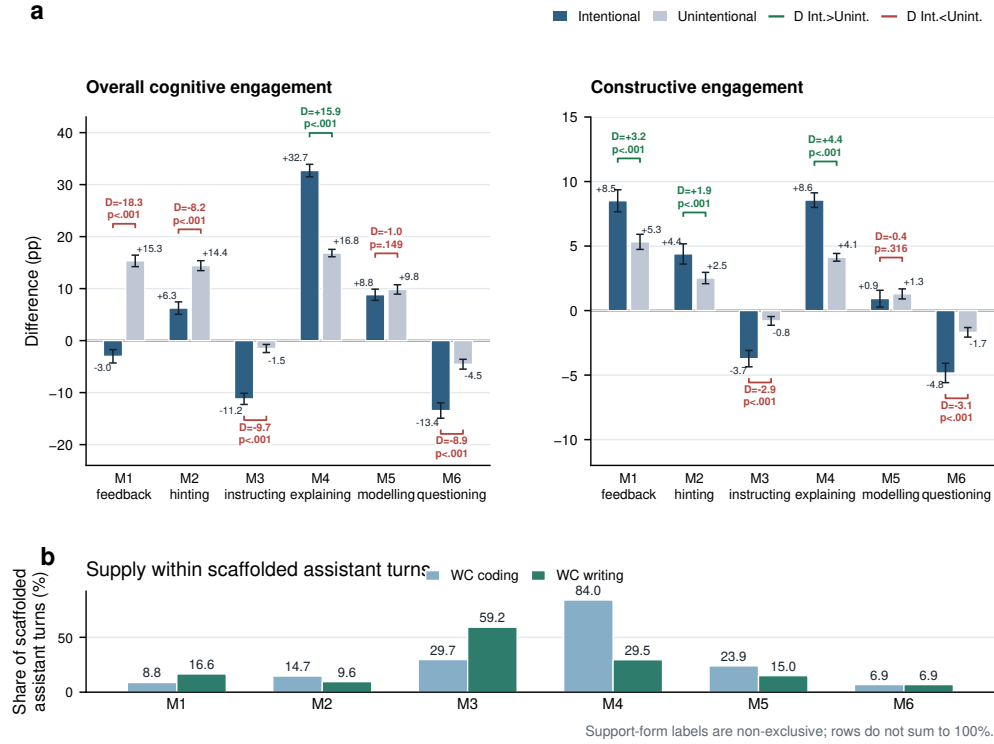


Fig. 4 Support form, user framing and support supply differentiate scaffolded interaction in WildChat. **a**, Engagement differences associated with each support form among scaffolded WildChat conversations, stratified by intentional versus unintentional learning orientation. Bars compare scaffolded conversations containing a support form with scaffolded conversations without that form; error bars show 95% confidence intervals, and brackets compare intentional and unintentional contrasts. **b**, Percentage of scaffolded assistant turns containing each support form by WildChat task setting. Support-form labels correspond to M1–M6 and are not mutually exclusive, so rows do not sum to 100%.

Assistant support-intent distinctions were less discriminating. Most scaffolded support in WildChat was cognitively oriented, whereas metacognitive and emotional support were much rarer. This reinforces the central role of support form—how support was delivered—rather than broad intent category.

Support supply made these form-level signatures consequential for task ecology (Fig. 4b). Coding-oriented scaffolded turns were dominated by explanation, with M4 appearing in 84.0% of scaffolded WildChat coding turns. Writing-oriented scaffolded turns were dominated by instructing, with M3 appearing in 59.2% of scaffolded WildChat writing turns. Users in different task settings therefore did not merely receive more or less scaffolding; they encountered different mixtures of support. In the sampled coding ecology, scaffolded support more often took the form of explanation, which was aligned with constructive sense-making. In the sampled writing ecology, scaffolded

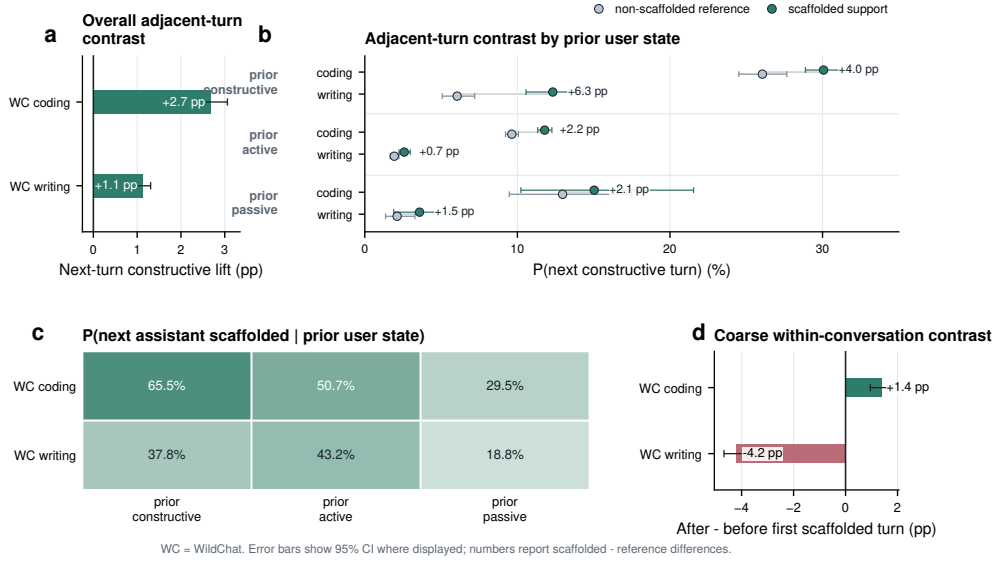


Fig. 5 Support-engagement coupling is local and state-dependent in WildChat. **a**, Next-turn constructive lift after scaffolded versus non-scaffolded reference assistant turns. **b**, Probability of a constructive next user turn by prior user state and support condition; annotations show scaffolded-minus-reference differences. **c**, Probability that the next assistant turn contains scaffolded support by prior user state. **d**, Constructive-engagement contrast after versus before the first scaffolded assistant turn. Error bars show 95% confidence intervals where displayed. WC, WildChat.

support more often took the form of instruction, which may support task completion but was less aligned with visible constructive elaboration.

Together, these results show why scaffolding presence alone is too coarse to characterize learning-oriented human-LLM interaction. Scaffolded support was associated with richer constructive participation at the conversation level, but that association was carried unevenly by support form, user framing and task-specific support supply. The implication is not that systems should simply provide more scaffolding, but that learning-oriented support needs to be matched to the user’s current framing and the task ecology: feedback and explanation may sustain users who are already working with ideas, whereas more directive forms may complete the task without eliciting further elaboration.

2.5 Support and constructive engagement are locally, not cumulatively, coupled

Having shown that scaffolding varied by support form, user framing and task-specific supply, we finally asked whether the support-engagement relationship was visible in the temporal ordering of turns. The clearest signal appeared at the adjacent-turn level. When assistant turns contained scaffolded support rather than falling in the non-scaffolded reference category, the next user turn was more likely to be constructive in both WildChat task settings (Fig. 5a). The lift was stronger in coding-oriented

conversations: +2.68 percentage points in WildChat coding (95% CI, 2.29 to 3.06). WildChat writing also showed a positive local lift (+1.13 points; 95% CI, 0.95 to 1.31). Thus, the most direct temporal evidence was local: scaffolded support was associated with constructive engagement in the next user turn, although the magnitude differed across task ecologies.

This local association was state-dependent rather than automatic. In WildChat coding, scaffolded support was followed by higher constructive engagement across prior constructive, active and passive user turns (Fig. 5b). WildChat writing showed the same positive direction across prior states, with the clearest lift after an already constructive user turn. The reverse ordering showed the same contingency: assistant scaffolding was more likely to follow constructive or active user turns than passive turns in coding-oriented conversations, while writing-oriented conversations also showed sensitivity to user state, with higher scaffolding after active or constructive turns than after passive turns (Fig. 5c). These patterns suggest that scaffolding was not simply an external input applied to otherwise stable conversations. User engagement shaped the support that appeared next, and scaffolded support in turn aligned with subsequent constructive engagement.

Coarser within-conversation contrasts were less consistent (Fig. 5d). Comparing constructive engagement after versus before the first scaffolded assistant turn yielded a small positive contrast in WildChat coding (+1.4 percentage points; 95% CI, 0.95 to 1.80) and a negative contrast in WildChat writing (-4.2 points; 95% CI, -4.68 to -3.82). We do not interpret the writing contrast as evidence that scaffolding reduces engagement; the first scaffolded turn may occur after difficulty, after an already constructive opening or at different points in the task trajectory. Instead, the mixed pattern argues against a simple cumulative-uplift account in which scaffolded support steadily raises constructive engagement over the remainder of a conversation. The more defensible conclusion is scale-specific: support and constructive engagement are locally and state-dependently coupled, rather than accumulating into a uniform upward trajectory.

Together, the temporal analyses sharpen the boundary of the support result. Scaffolded support was associated with constructive participation, but the strongest evidence was local, state-dependent and task-specific. Everyday human-LLM interaction therefore looks less like a one-way instructional pipeline and more like a coupled interaction process: users' current engagement shapes the support they receive, and scaffolded support is locally associated with subsequent opportunities for constructive engagement. Learning-oriented engagement in everyday LLM use therefore depends not on exposure to support alone, but on situated sequences in which task ecology, user framing, support form and user state jointly shape whether people continue working constructively with ideas.

3 Discussion

This study identifies everyday human-LLM interaction as a middle space between ordinary tool use and designed instruction. In naturalistic coding and writing conversations, learning-oriented engagement was recurrent, but its constructive form was

selective. Users often used, applied or followed up on assistant responses; deeper constructive engagement emerged when the exchange preserved reasons and opportunities for users to keep working with ideas—through explicit learning framing, task affordances that made testing and revision salient, and support that left room for user evaluation rather than simply completing the task. Assistant scaffolding was therefore not informative as a uniform support exposure. It was associated with richer constructive participation when its form, timing and surrounding user state supported continued sense-making: feedback and explanation aligned more clearly with constructive engagement than more directive or question-based moves, and temporal evidence was strongest in adjacent-turn coupling rather than cumulative uplift. Together, these findings suggest that informal learning in everyday LLM use is a situated interactional achievement, not a generic consequence of AI access: it emerges when users and assistants maintain a shared space for testing, revising and extending understanding while pursuing practical goals.

The results suggest that constructive engagement in everyday AI use is ecological. It is not simply a property of an individual user, nor a direct consequence of a model producing educational content. Instead, it depends on how the user’s goal, the task ecology and the unfolding exchange configure opportunities for sense-making. Explicit learning framing amplified constructive engagement, but constructive participation also appeared in task-oriented conversations where learning was incidental. This matters because informal learning rarely announces itself as a formal learning episode. It often appears when a practical task exposes uncertainty, invites comparison or requires revision. Coding and writing illustrate this point as task ecologies rather than as isolated topics. Some tasks make breakdowns, constraints and tests more visible, while others allow users to accept or apply a revision without articulating the rationale. Similarly, sustained exchanges should not be treated as evidence of learning progression by themselves; they are opportunity structures in which users can return to, interrogate and reshape the assistant’s response. Everyday LLM use should therefore not be analysed as a single behavioural environment. Different tasks organize different possibilities for users to ask, test, refine and extend ideas.

The support results further suggest that scaffolding is best understood as an interactional condition rather than a treatment-like exposure. Conversations containing scaffolded support showed richer constructive participation, greater post-answer depth and, in coding-oriented conversations, greater persistence after observable breakdowns. These patterns indicate that scaffolding is part of the conversational environment in which constructive participation becomes more likely to appear. At the same time, scaffolded support was not randomly assigned. More difficult tasks, more motivated users or already constructive exchanges may elicit scaffolding while also producing constructive follow-up. The point is therefore not that exposure to scaffolding causes durable learning gains. Rather, the study identifies structured associations in naturalistic interaction and shows where they become visible: in conversations where support and user participation are mutually organized around continued work with ideas.

The most actionable insight is that learning-oriented support is not a matter of adding more scaffolding in general. The broad label of scaffolding concealed important differences in support form. Feedback-like support and explanation were most clearly

aligned with constructive engagement, whereas instructing and questioning were less aligned with visible elaboration. This distinction suggests that different support forms leave different roles for the user. Feedback responds to a user’s prior attempt and can create an occasion for revision, evaluation or extension. Explanation can make mechanisms visible and give users material for testing their understanding. By contrast, detailed instructions can make the next step executable without requiring the user to articulate why it works, and questions in everyday assistant use may function as information gathering or clarification requests rather than carefully designed prompts for reflection. This does not mean that instruction or questioning are unhelpful. They may be appropriate when users need direct task support, when information is missing or when the cost of continued exploration is high. But they illustrate why support quality for informal learning cannot be judged only by whether a response is pedagogical in form. The relevant question is whether the support preserves a productive role for the user in the work of understanding.

The temporal results clarify a division of labour between AI-mediated informal learning and formal instruction. Everyday LLM interactions repeatedly create local moments in which support and user engagement align, but those moments do not automatically accumulate into a coherent learning trajectory. This pattern does not mean that everyday AI use should be judged by whether it reproduces formal instruction. Rather, it points to a complementary role. Formal instruction is designed to accumulate learning over time through curricular sequencing, shared standards, assessment, accountability and opportunities to abstract beyond the immediate task. Everyday AI use is more episodic and situated: it enters moments when users are confused, blocked, comparing alternatives, repairing a failed attempt or adapting knowledge to a concrete problem. Its value is therefore not to replace formal education, but to extend learning into the spaces between and around formal learning episodes, where people encounter unfamiliar concepts while doing real work. The temporal findings show both the promise and the limitation of this role. LLM interaction can create situated opportunities for constructive engagement, but those opportunities require additional design support if they are to become durable knowledge, transferable skill or cumulative expertise. Future assistants may therefore need to bridge informal and formal learning functions: preserving the immediacy and personalization of point-of-need support, while helping users connect repeated local episodes into longer arcs of reflection, consolidation and skill development.

This complementarity suggests a design principle for personal AI assistants that support lifelong learning: they should not simply imitate formal tutors or add instructional language, but should manage the cognitive responsibility left to the user across time. At the local scale, support should be calibrated to the user’s state and the task affordance: direct help may be appropriate when completion is the immediate goal, whereas explanation, feedback or partial cues are more useful when the user is trying to diagnose, evaluate or revise an idea. Equally important is restraint, because too much instruction can replace the user’s own sense-making rather than support it. At the longitudinal scale, the assistant must do more than choose the right support form in a single exchange. Because local support–engagement alignments do not automatically accumulate, future systems should help users connect repeated episodes across

time: recognizing recurring uncertainties, linking current problems to prior explanations, and turning repeated breakdowns into occasions for reflection and consolidation. In this sense, the distinctive promise of lifelong AI assistance is not to convert everyday work into formal instruction, but to give continuity, memory and personalization to the informal learning that happens between formal learning episodes. Together with formal instruction, such systems could support a more continuous and personalized learning ecosystem in which schools and curricula provide structure, standards and shared progression, while personal AI assistants carry learning into everyday practice, helping people transform formal knowledge into situated judgment, transferable skill and cumulative understanding.

Several boundaries define the scope of these claims. The analyses use naturalistic logs, so they identify structured associations rather than causal effects. Scaffolded support was not assigned, and conversations that elicited support may also have involved more complex tasks, more motivated users or prior engagement. The data also capture interactional episodes rather than long-term learning histories: they do not show whether repeated use leads individual users to retain knowledge, transfer skills or become more independent over time. In addition, engagement labels are behavioural proxies. Constructive engagement marks observable elaboration, testing, revision or extension, but it is not a direct measure of durable learning outcomes. Finally, the corpus covers large public WildChat sample covering two common task ecologies, coding and writing; other domains, populations and deployment contexts may organize different opportunities for engagement and support. Future work should therefore combine large-scale log analysis with controlled interventions, longitudinal user-level designs and independent measures of knowledge gain or skill transfer.

Within these boundaries, the study makes everyday AI-supported informal learning empirically visible. It does not show that LLM support causes learning; instead, it shows where learning-oriented engagement appears, how its constructive form is shaped by task ecology and user framing, and why assistant support must be understood through form, timing and local user state rather than through scaffolding presence alone. This reframes everyday human–LLM interaction as a middle space between ordinary tool use and formal instruction. Together with formal instruction, such systems could support a more continuous and personalized learning ecosystem in which curricula provide structure, standards and shared progression, while personal AI assistants extend learning into everyday practice, helping learners transform situated problem solving into durable understanding, transferable expertise and lifelong learning agency.

4 Methods

4.1 Study design

This study is an observational analysis of naturalistic WildChat human–LLM conversations across two everyday task ecologies, coding and writing. It asks whether learning-oriented engagement is visible in ordinary use, where constructive engagement is concentrated and how assistant scaffolding is ordered with user engagement across conversation-level and adjacent-turn timescales. The design is descriptive and

associational. It does not estimate randomized exposure effects, causal instructional efficacy or durable learning outcomes.

The empirical sequence mirrors the Results section. We first estimate the prevalence and composition of user engagement, then examine how constructive engagement varies by user framing, task ecology and interaction depth. We next test whether scaffolded assistant support marks conversations with richer constructive participation, separate scaffolded support by support form and analyse local temporal coupling between assistant support and user engagement. Dataset and task ecology are treated as substantive dimensions of the interaction environment rather than nuisance variables to be averaged away.

4.2 Data sources and analytic samples

The analytic corpus contains 71,412 WildChat conversations and 577,225 total turns, including 287,778 user turns and 289,447 assistant turns. The two task settings are WildChat coding and WildChat writing. WildChat coding contributes 31,878 conversations with 124,073 user turns and 124,623 assistant turns. WildChat writing contributes 39,534 conversations with 163,705 user turns and 164,824 assistant turns.

The two WildChat task settings were analysed with a common Level 1–4 annotation and analysis framework so that prevalence, contextual variation, support–engagement association and temporal ordering were computed from the same construct definitions. The manuscript reports the refreshed production outputs after the latest user-turn update pass rather than earlier pre-update batches. Supplementary Section A.1 describes the corpus construction and refresh provenance in more detail.

4.3 Conversation metadata and task ecologies

Each conversation is represented as an ordered sequence of user and assistant turns together with available metadata. The main contextual variables are data source, task domain, learning intent, topic and language. Coding and writing were selected because they are common everyday LLM use cases but differ in their interactional affordances: coding often involves debugging, error diagnosis and iterative testing, whereas writing often involves drafting, revision and directive editing. Learning intent distinguishes conversations explicitly oriented toward understanding, exploration or learning from conversations in which learning-oriented engagement may be incidental to task completion.

Behavioural depth is measured from the number and ordering of turns in the conversation. Depth is used descriptively, as a marker of interactional opportunity and sustained exchange, not as an independent manipulation. Post-answer depth is measured as the number of turns after the first assistant response.

4.4 Turn-level constructs

The unit of annotation is the conversational turn. User and assistant turns are labelled with different constructs because they play different roles in the interaction. User turns are labelled for behavioural, emotional and cognitive engagement. The

main analyses emphasize cognitive engagement and its depth levels. Passive engagement (P) captures reception or acknowledgement without substantial transformation. Active engagement (A) captures task-relevant action, follow-up or manipulation of supplied information. Constructive engagement (C) captures elaboration, revision, testing, extension or transformation of ideas. Constructive engagement is treated as the strictest behavioural proxy for learning-oriented participation available in these logs, not as a direct measure of knowledge gain.

Assistant turns are first distinguished as S1 or S2. S1 denotes a non-scaffolded reference response, including straightforward task fulfilment or stand-alone information provision. S2 denotes scaffolded support: assistant behaviour that structures, regulates or redirects the user’s process while leaving some cognitive work with the user. This definition follows the scaffolding tradition, especially contingency, transfer of responsibility and potential fading [11, 12, 14]. Scaffolded turns can also receive intention labels (I1 metacognitive, I2 cognitive and I3 affective support) and non-mutually exclusive means labels (M1 feedback, M2 hinting, M3 instructing, M4 explaining, M5 modelling and M6 questioning). Operational examples are provided in Supplementary Table C1.

4.5 Annotation workflow and verification

The annotation workflow combined theory-grounded codebook development, prompt iteration, parser checks, human review and production-scale LLM-assisted labelling. Prompt and post-processing rules were first developed on validation batches to reduce common failure modes, including over-labelling brief acknowledgements as active engagement and misclassifying directive assistant responses as scaffolding. After the pipeline stabilized, the same annotation logic was applied to the WildChat analytic corpus. All examples reproduced in the manuscript are de-identified and paraphrased rather than copied from raw logs.

Human verification was used as a quality-control input rather than as a separate source of outcome measurement. The final coding validation rerun combines three review rounds, with 658 task turns and 643 turns after deduplication. The final writing validation used 210 reviewed conversations. Agreement was summarized with per-label F1, Matthews correlation coefficient and Gwet’s AC1 because the labels are sparse, prevalence-imbalanced and not mutually exclusive in all cases. Agreement was strongest for constructive, passive, emotional and support-form labels; boundary-heavy labels such as writing A and S1 were less stable. Supplementary Table C2 reports the per-label agreement metrics, and Supplementary Section A.4 describes the final user-turn update check.

4.6 Outcome construction

The primary user-side outcome is constructive engagement. At the turn level, this is a binary indicator for whether the user turn is labelled C. At the conversation level, we use both the count of constructive user turns and an indicator for whether a conversation contains at least one constructive user turn. Constructive ratios are defined as constructive user turns divided by user turns.

The primary assistant-side exposure is scaffolded support. At the turn level, this is the distinction between S2 scaffolded support and S1 non-scaffolded reference response. At the conversation level, we use an indicator for whether a conversation contains at least one S2 assistant turn. Support-form analyses condition on scaffolded turns and compare conversations containing each means label with scaffolded conversations that do not contain that label. Because means labels can co-occur, these are descriptive support-form contrasts rather than mutually exclusive treatment comparisons.

4.7 Statistical analyses

Descriptive analyses estimate engagement prevalence, scaffolding prevalence, passive/active/constructive composition, emotional-engagement rates and behavioural-depth summaries within each WildChat task setting. Context analyses compare intentional versus unintentional conversations and coding versus writing settings. Depth analyses group conversations into behavioural-depth buckets and estimate the probability that a conversation contains at least one constructive turn.

Conversation-level associations are analysed by comparing conversations with and without at least one scaffolded assistant turn. Constructive-turn counts are modelled with Poisson generalized linear models. Binary outcomes, including whether a conversation contains any constructive turn, whether the next user turn is constructive and whether the next assistant turn is scaffolded, are modelled with logistic regression or reported as conditional probability contrasts, depending on the estimand. Adjusted models include observed covariates appropriate to the analysis level, including learning intent, task domain or topic indicators, turn counts, emotional ratios, error-related variables and turn index. Supplementary Tables C3–C5 summarize the main model families and temporal estimands.

Temporal analyses use two complementary scales. Adjacent-turn analyses compare the probability that the next user turn is constructive after scaffolded versus non-scaffolded assistant turns and the probability that the next assistant turn is scaffolded after different prior user states. Coarser within-conversation analyses compare constructive engagement before versus after the first scaffolded assistant turn and first-half versus second-half summaries where available. The adjacent-turn analyses test local ordering; the coarser analyses test whether there is evidence for longer-span accumulation.

4.8 Robustness, privacy and LLM use

WildChat robustness analyses repeat the main patterns across user-facing model snapshots and coarse model families. These checks assess whether the headline patterns are reducible to a single WildChat model label. They are not interpreted as model-causal comparisons, because model labels are not experimentally assigned and may be confounded with time, user population, task mix and logging context.

The analyses are bounded by the observational design and by the nature of the labels. They support claims that everyday human–LLM conversations contain measurable learning-oriented engagement; that constructive engagement varies by intent,

task ecology and depth; that assistant scaffolding is associated with richer constructive participation; and that support and engagement show local reciprocal ordering. They do not support claims that scaffolded support causes durable learning gains, that all constructive engagement reflects actual knowledge change or that the labelled conversational states perfectly measure learning itself.

LLMs were used as objects of study and as annotation tools. Production labels were generated with LLM-assisted workflows and then checked through parser rules, targeted update scripts and human validation. The authors also used LLM-based tools to assist with code drafting, debugging, analysis workflow organization and language editing. LLM-generated assistance was not used as a substitute for human verification of the validation samples or for interpretation of the statistical results.

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Supplementary information

Supplementary information is provided below.

Appendix A Supporting methods

A.1 Corpus construction and analytic settings

The analyses use two WildChat task settings: coding-oriented conversations and writing-oriented conversations. Coding and writing were selected because both are common everyday LLM use cases, yet they differ in the opportunities they create for visible learning-oriented engagement. Coding conversations often contain error messages, debugging attempts, execution failures and iterative repair. Writing conversations more often involve drafting, rewriting, shortening, tone adjustment and direct editing.

The analytic corpus contains 71,412 WildChat conversations. WildChat contributes 31,878 coding conversations and 39,534 writing conversations. All reported analyses use the refreshed Level 1–4 outputs after the latest user-turn update pass.

For each conversation, the pipeline retained the ordered turn sequence, role labels, available topic and language metadata, learning-intent indicators and model metadata where available. The model metadata are used only for robustness checks in WildChat because model assignment in naturalistic public conversations is observational and may reflect time, user population, task mix and logging context.

A.2 Annotation ontology

The codebook translates learning-science constructs into role-specific behavioural signatures. User turns are labelled for engagement. Passive engagement (P) marks acknowledgement, reception or minimal continuation without visible transformation. Active engagement (A) marks task-relevant action or follow-up, such as asking for a reformatted answer or applying supplied information. Constructive engagement (C) marks deeper work with ideas, such as testing an implication, revising a hypothesis, articulating a constraint, transferring a mechanism or extending an explanation. Emotional engagement is labelled separately because affective expressions were rare and not the main analytic outcome.

Assistant turns are labelled for support. S1 denotes a non-scaffolded reference response: the assistant provides the requested information or deliverable without materially structuring the user’s process. S2 denotes scaffolded support: the assistant structures, regulates, diagnoses or redirects the user’s work while leaving some responsibility with the user. Scaffolded turns can also receive support-intent labels and support-means labels. Intention labels distinguish metacognitive support (I1), cognitive support (I2) and affective support (I3). Means labels distinguish feedback (M1), hinting (M2), instructing (M3), explaining (M4), modelling (M5) and questioning (M6). Means labels are non-mutually exclusive because a single assistant turn can both explain a mechanism and provide an example or hint. Supplementary Table C6 reports the support-means supply profile across the two WildChat task settings.

A.3 LLM-assisted annotation pipeline

The annotation pipeline combined prompt-based labelling with deterministic parser checks and targeted post-processing. During development, prompts were iterated against validation examples and disagreement cases. Common failure modes included treating brief acknowledgements as active engagement, treating any long assistant answer as scaffolding, collapsing directive instructions into scaffolding without checking whether responsibility remained with the user, and over-labelling sparse support means.

Production labels were generated at turn level. Parser checks enforced the expected label schema, normalized label variants and flagged missing or malformed outputs. Post-processing rules were used only when a recurring error pattern could be specified transparently, such as a narrow class of user-turn false positives. Prompt versions, run metadata, disagreement files and update outputs were retained with the analysis artifacts so that validation cases can be traced without reproducing raw conversation text in the manuscript.

A.4 Human verification and update provenance

Human verification was used to quantify label reliability and identify boundary cases. Coding validation combined three review rounds, with 658 task turns and 643 turns after deduplication. Writing validation used 210 reviewed conversations. Because many labels are sparse or prevalence-imbalanced, agreement is reported with per-label F1, Matthews correlation coefficient and Gwet’s AC1 rather than with a single global agreement statistic.

The final user-turn check contained 100 WildChat coding examples and 100 WildChat writing examples reviewed after the update pass. These examples were used to identify residual false positives and calibrate patch-style update filters, especially for Active and Passive user-turn labels. The last WildChat user-turn update pass scanned 31,878 coding conversations and 39,534 writing conversations, applying 1,815 coding updates and 21 writing updates. The main manuscript reports the refreshed labels produced by this final pass.

A.5 Outcome construction

At the turn level, the focal user-side outcome is a binary indicator for constructive engagement. At the conversation level, we use both the count of constructive user turns and an indicator for whether at least one constructive user turn occurs. Constructive ratios divide constructive user turns by total user turns. Cognitive engagement ratios include passive, active and constructive cognitive engagement.

At the assistant side, scaffolded support is represented by the **S2** label. Conversation-level scaffolding presence indicates whether at least one assistant turn in the conversation is labelled **S2**. Support-form analyses condition on conversations or turns with scaffolded support and then compare the presence versus absence of each means label. These comparisons are descriptive because means labels can co-occur and because conversations are not randomly assigned to receive a support form.

Behavioural depth is computed from turn counts and turn ordering. Post-answer depth counts turns after the first assistant response. Persistence after failure is analysed only for coding-oriented conversations because debugging failures, error messages and failed-execution contexts are more directly observable in coding than in writing.

A.6 Statistical model specification

The main analyses use simple estimands matched to the claim being made. Descriptive prevalence analyses report proportions within WildChat task settings. Context analyses compare learning intent, task ecology and depth groups. Conversation-level support associations use Poisson generalized linear models for constructive-turn counts and logistic models for binary constructive-engagement outcomes. Adjacent-turn analyses use assistant-to-user and user-to-assistant turn pairs to estimate local conditional probabilities.

Covariates are included when they correspond to observed compositional differences at the analysis level. Depending on the analysis, these include learning intent, topic or task indicators, turn counts, emotional ratios, error-related variables and turn index. Adjusted estimates are interpreted as measured-covariate-adjusted associations. They do not remove unobserved user motivation, prior expertise, task difficulty or time-varying conversational state.

A.7 Temporal scale

The temporal analyses deliberately separate adjacent-turn ordering from coarser within-conversation summaries. The adjacent-turn analyses ask whether the immediately following user turn is more likely to be constructive after scaffolded support than after a non-scaffolded reference response, and whether that contrast depends on the prior user state. The reverse adjacent-turn analysis asks whether the assistant is more likely to provide scaffolded support after constructive, active or passive user turns.

The coarser analyses ask a different question: whether constructive engagement is higher after the first scaffolded assistant turn than before it, and whether constructive engagement increases from the first to the second half of a conversation. These are not treatment-effect estimates. The first scaffolded turn may occur precisely because the user is struggling, because a difficult task has emerged or because the exchange is already developing. The main temporal claim is therefore scale-specific: the clearest evidence concerns local, state-conditioned coupling rather than cumulative uplift.

Appendix B Robustness and interpretive boundaries

B.1 WildChat model-snapshot robustness

WildChat includes user-facing model metadata that make it possible to repeat major descriptive patterns across model snapshots and coarse model families. These analyses preserved the main directional patterns reported in the main text: coding generally showed higher cognitive engagement than writing; intentionality gaps were visible across most user-facing models; scaffolded conversations were generally more constructive; and adjacent-turn contrasts were more interpretable than coarser

within-conversation contrasts. Supplementary Figures D3–D7 report these WildChat model-snapshot checks.

These checks are used as robustness analyses rather than model-causal comparisons. WildChat model labels are not experimentally assigned. They may be confounded with calendar time, user population, task mix, interface changes and logging context. For this reason, the manuscript does not claim that a specific model family caused the observed engagement differences.

B.2 Label uncertainty

The label scheme intentionally separates top-level constructs from finer support forms. Top-level labels such as constructive engagement and scaffolded support have clearer behavioural anchors. Fine-grained labels, especially sparse or boundary-heavy labels, are less stable. This is why the main claims rely most heavily on constructive engagement, scaffolded support presence, support-form patterns that are consistent across settings and temporal contrasts that can be expressed with simple conditional probabilities.

B.3 Selection into support

The most important threat to causal interpretation is selection into support. More complex tasks, more motivated users or already-developing exchanges may both elicit scaffolded support and produce constructive follow-up. Conversation-level adjusted models reduce measured compositional differences but cannot remove unobserved selection. Adjacent-turn analyses reduce the temporal window and condition on local state, but they still cannot identify randomized exposure effects. The safest interpretation is therefore local, state-dependent coupling between assistant support and user engagement.

B.4 Privacy-preserving reporting

The study analyses naturalistic conversations, so the manuscript avoids reproducing raw conversation text. Operational examples are de-identified and paraphrased to illustrate decision rules without exposing user content. Aggregate statistics, model summaries and validation metrics are reported in the manuscript and supplementary tables.

Appendix C Supporting tables

Table C1 Operational examples for turn-level labels. Examples are de-identified and paraphrased from validation review patterns; they illustrate the decision rule rather than reproducing raw conversation text.

Construct	Paraphrased turn	Annotation rationale
P passive user engagement	"Thanks, that makes sense."	Minimal uptake or acknowledgement without new reasoning, evaluation or task transformation.
A active user engagement	"Can you convert that answer into a command I can run in my setup?"	Task-relevant follow-up or manipulation of supplied information, but without elaborating or testing an idea.
C constructive user engagement	"If the cache expires before the asynchronous callback returns, would the same race condition still happen?"	The user tests an implication, boundary condition or transfer of the prior explanation.
S1 non-scaffolded reference	"Here is the requested function/email draft."	The assistant primarily completes the requested deliverable without regulating the user's process.
S2 scaffolded support	"First isolate the failing step; that will show whether the issue is parsing or state management."	The assistant structures the user's process and leaves diagnostic or revision work with the user.
M1 feedback	"Your diagnosis is close; the bug is in the index offset, not the loop boundary."	Evaluative feedback on the user's prior attempt or understanding.
M2 hinting	"Check the parser log around the first malformed token before changing the model."	A cue or partial path that helps the user proceed without fully resolving the task.
M3 instructing	"First split the condition into two branches, then add a regression test."	Explicit steps or strategy that organizes the user's work.
M4 explaining	"The exception occurs because the event loop is already running, so a nested call cannot acquire it."	A rationale or mechanism that clarifies why the issue occurs.
M5 modeling	"A small pattern you can adapt is: validate input, normalize it, then pass it to the writer."	A sample approach intended to be emulated or adapted rather than merely copied.
M6 questioning	"Which constraint matters more here: preserving tone or shortening the paragraph?"	A question that advances problem solving or reflection.

Table C2 Human–human agreement for labels used in the main analyses. Coding uses the latest three-round validation rerun over 658 task turns, 643 after deduplication; writing uses the latest writing-210 best-pipeline validation. F1, Matthews correlation coefficient (MCC) and Gwet’s AC1 are reported per binary label. Coding MCC and AC1 are recomputed from the exported support and positive-count fields; writing metrics are from the exported agreement report. Rare degenerate labels with no positive cases in a validation set are omitted.

Domain	Role	Label	F1	MCC	Gwet AC1	Support
Coding	User	Active (A)	0.75	0.72	0.87	315
Coding	User	Constructive (C)	0.85	0.82	0.90	315
Coding	User	Passive (P)	0.88	0.87	0.96	315
Coding	User	Emotional (E)	0.83	0.82	0.97	315
Coding	Assistant	Non-scaffolded reference (S1)	0.78	0.73	0.88	328
Coding	Assistant	Scaffolded support (S2)	0.87	0.69	0.70	328
Coding	Assistant	Feedback (M1)	0.90	0.89	0.95	328
Coding	Assistant	Hinting (M2)	0.79	0.75	0.89	328
Coding	Assistant	Instructing (M3)	0.81	0.77	0.92	328
Coding	Assistant	Explaining (M4)	0.86	0.79	0.84	328
Coding	Assistant	Modeling (M5)	0.84	0.79	0.87	328
Coding	Assistant	Questioning (M6)	0.82	0.81	0.95	328
Writing	User	Active (A)	0.38	0.38	0.69	90
Writing	User	Constructive (C)	0.89	0.83	0.86	90
Writing	User	Passive (P)	0.80	0.81	0.99	90
Writing	User	Emotional (E)	0.67	0.65	0.96	90
Writing	Assistant	Non-scaffolded reference (S1)	0.46	0.43	0.86	120
Writing	Assistant	Scaffolded support (S2)	0.97	0.80	0.93	120
Writing	Assistant	Feedback (M1)	0.89	0.80	0.80	120
Writing	Assistant	Hinting (M2)	0.82	0.70	0.71	120
Writing	Assistant	Instructing (M3)	0.71	0.67	0.86	120
Writing	Assistant	Explaining (M4)	0.84	0.70	0.70	120
Writing	Assistant	Modeling (M5)	0.94	0.91	0.95	120
Writing	Assistant	Questioning (M6)	0.96	0.96	0.99	120

Table C3 Analytic model families used in the main analyses. The table summarizes descriptive and conversation-level model families. Adjusted models are interpreted as covariate-adjusted associations in naturalistic data, not as causal treatment-effect estimates.

Analysis family	Unit and outcome	Main contrast or model	Scope
Engagement prevalence	User turns and conversations; cognitive, constructive and emotional engagement indicators.	Descriptive proportions by WildChat task setting, task ecology and learning intent.	Establishes where learning-oriented engagement is visible; no causal contrast is implied.
Engagement composition	Cognitively engaged user turns; passive, active and constructive depth levels.	Composition of cognitive engagement within each WildChat task setting.	Motivates constructive engagement as the focal strict outcome while preserving the broader cognitive-engagement hierarchy.
Interaction depth	Conversations; at least one constructive user turn.	Probability of constructive engagement across behavioural-depth buckets.	Depth is interpreted as interactional opportunity and task development, not as a manipulated exposure.
Scaffolding presence	Conversations; constructive-turn count and any constructive turn.	Poisson generalized linear models for counts and logistic models for binary outcomes, comparing conversations with versus without scaffolded assistant turns.	Adjusted estimates reduce measured compositional differences but remain associational because support is selected into by task difficulty and user state.

Table C4 Additional analytic model families and robustness checks. These analyses qualify the main associations by support form, temporal scale and available WildChat model metadata.

Analysis family	Unit and outcome	Main contrast or model	Scope
Support-form associations	Scaffolded conversations; cognitive or constructive engagement ratios.	Mean differences for conversations containing a given support-means label versus scaffolded conversations without that label, stratified by learning intent.	Descriptive because means labels are non-exclusive and can co-occur in the same assistant turn.
Local temporal coupling	Adjacent assistant–user or user–assistant turn pairs; next constructive user turn or next scaffolded assistant turn.	Conditional probability contrasts around adjacent turns; detailed estimands are listed in Supplementary Table C5.	Interpreted as local ordering and state-dependent coupling, not as randomized support effects.
Coarse temporal contrasts	Conversations containing scaffolded support; constructive engagement before and after the first scaffolded turn.	Mean after-minus-before contrast and first-half versus second-half robustness summaries where available.	Tests longer-span accumulation patterns, but the first scaffolded turn may be selected by difficulty, uncertainty or an already developing exchange.
WildChat model robustness	WildChat conversations or turns; main prevalence, association and temporal outcomes.	Repetition across user-facing model snapshots or coarse model families.	Robustness summaries only; model labels are observational and may be confounded with time, user mix and task mix.

Table C5 Temporal estimands used to separate local coupling from coarse accumulation. The analyses intentionally combine adjacent-turn and longer-span summaries because the theoretical claim concerns temporal scale. Adjacent-turn contrasts test local ordering; within-conversation contrasts test whether there is evidence for broader accumulation after scaffolding appears.

Estimand	Definition	Reported in main text	Scope
Overall adjacent-turn lift	Difference in the probability that the next user turn is constructive after scaffolded support versus after a non-scaffolded reference response, estimated within each WildChat task setting.	Figure 5a	Local association between assistant support type and immediately following constructive engagement; not a randomized support effect.
Prior-state adjacent contrast	Difference in next-turn constructive probability after scaffolded support versus reference response, separately for prior constructive, active and passive user states.	Figure 5b	Tests whether local support–engagement coupling depends on the user’s immediately preceding state.
Reverse state-conditioned support	Probability that the next assistant turn is scaffolded, estimated after constructive, active and passive user turns.	Figure 5c	Describes how assistant scaffolding supply varies with the user’s current engagement state.
Around-first-scaffold contrast	Mean constructive engagement after the first scaffolded assistant turn minus mean constructive engagement before that turn, among conversations containing scaffolded support.	Figure 5d	Coarse before–after contrast; the first scaffolded turn may be selected by difficulty, uncertainty or an already developing exchange.
First-half versus second-half contrast	Constructive engagement in the second half of a conversation minus constructive engagement in the first half, reported where available in WildChat.	Results text	A longer-span robustness lens; increases do not identify scaffolding as the cause of later engagement.

Table C6 Support-supply profile within scaffolded assistant turns. Values show the percentage of scaffolded assistant turns containing each scaffolding-means label. Means labels are not mutually exclusive, so rows do not sum to 100%. Coding-oriented scaffolded turns are dominated by explanation-heavy support, whereas writing-oriented scaffolded turns are dominated by instructing support.

Setting	M1 Feedback	M2 Hinting	M3 Instructing	M4 Explaining	M5 Modelling	M6 Questioning
WildChat coding	8.8	14.7	29.7	84.0	23.9	6.9
WildChat writing	16.6	9.6	59.2	29.5	15.0	6.9

Appendix D Supporting figures

The supporting figures report extended temporal checks and WildChat model-snapshot robustness analyses. They are included to make the main claims auditable without shifting the manuscript’s emphasis away from the two WildChat task settings. Model-snapshot figures are descriptive robustness checks because WildChat model labels are observational rather than experimentally assigned.

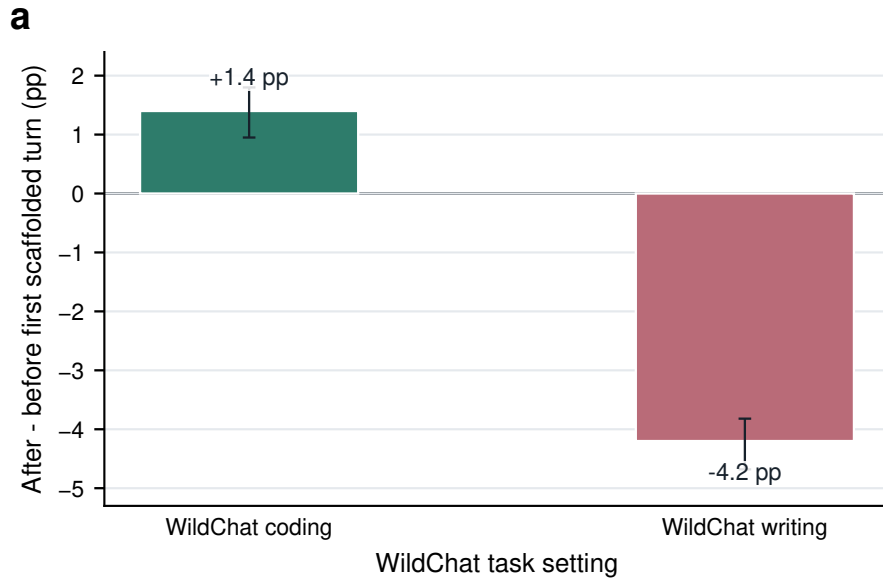


Fig. D1 Coarse around-first-scaffold contrasts provide limited evidence for cumulative improvement. Bars show the difference in constructive engagement after versus before the first scaffolded assistant turn within WildChat conversations containing scaffolded support. The contrasts are mixed across the two task settings, supporting the main-text interpretation that the clearest temporal evidence concerns local adjacent-turn coupling rather than broad cumulative uplift.

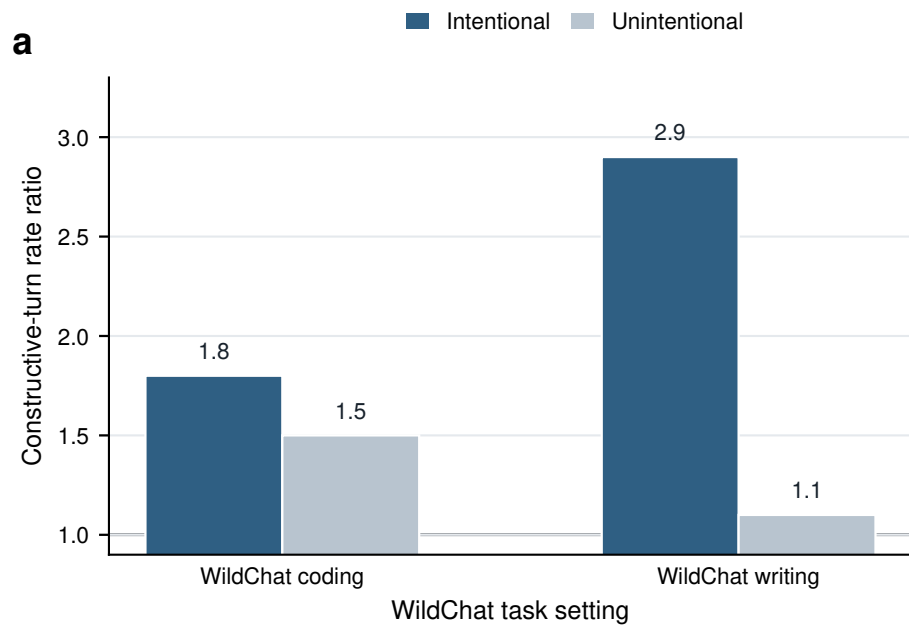


Fig. D2 Scaffolded-support associations vary by user framing. Stratified model estimates compare the association between scaffolded-support presence and constructive participation in intentional versus unintentional conversations. The figure is interpreted as descriptive moderation rather than evidence that intent causally changes support effectiveness, because user framing, task complexity and persistence are not randomly assigned.

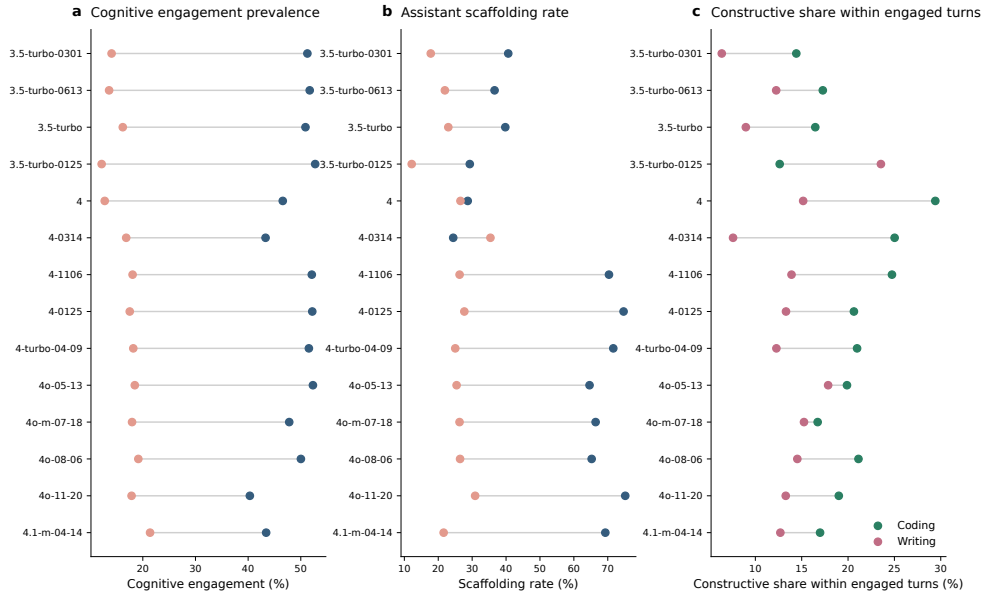


Fig. D3 WildChat model-snapshot prevalence profiles preserve the main domain contrast. Across user-facing WildChat model snapshots, coding-oriented conversations generally show higher cognitive engagement and higher assistant scaffolding rates than writing-oriented conversations. The model labels are observational and may reflect time, user mix and traffic composition, so the figure is used as a robustness check rather than a model-causal comparison.

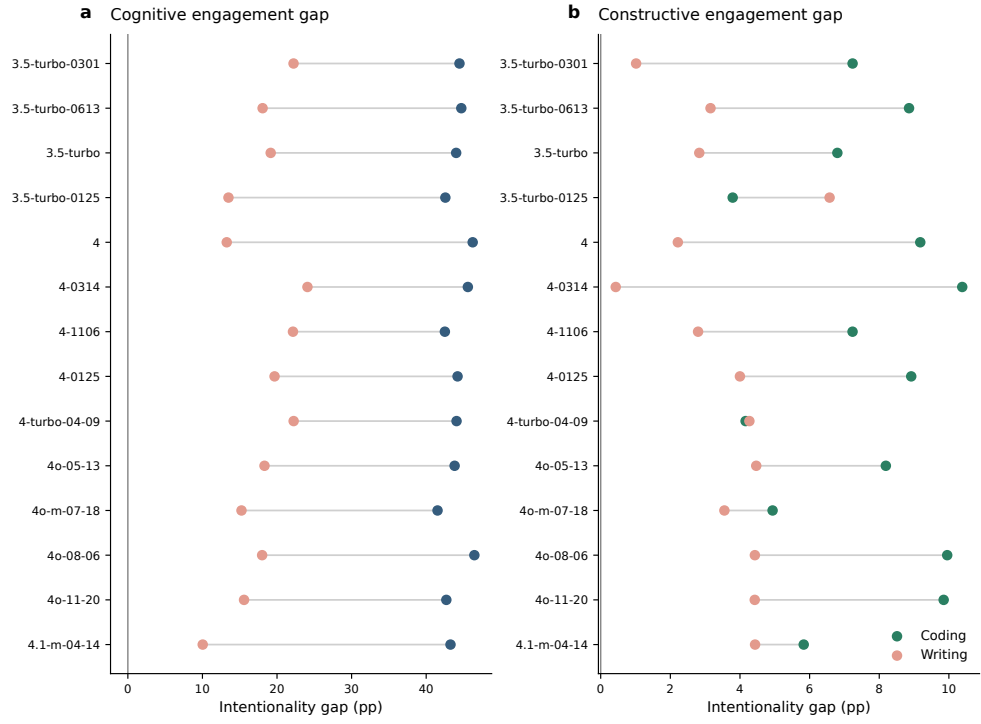


Fig. D4 Intentionality gaps are visible across WildChat model snapshots. The figure reports model-specific differences between intentional and unintentional conversations in cognitive and constructive engagement, separately for coding and writing. Most model snapshots preserve the directional pattern that intentionally framed conversations contain richer learning-oriented participation.

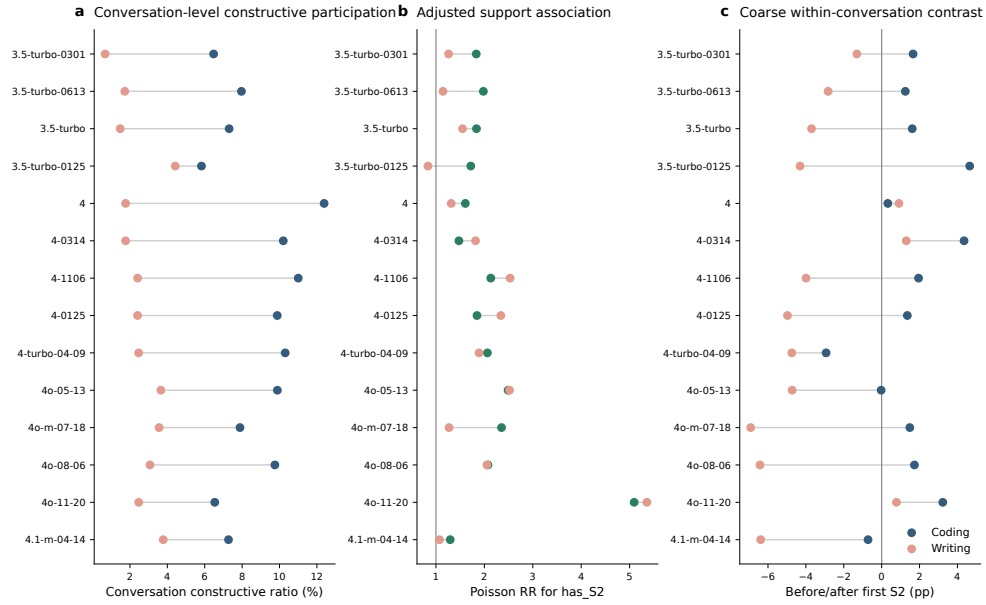


Fig. D5 Conversation-level support associations are broadly positive across WildChat model snapshots. The figure repeats major conversation-level analyses by user-facing WildChat model snapshot, including constructive participation, adjusted support associations and coarse before–after contrasts around the first scaffolded assistant turn. The pattern supports the distinction between stable conversation-level associations and more mixed coarse temporal contrasts.

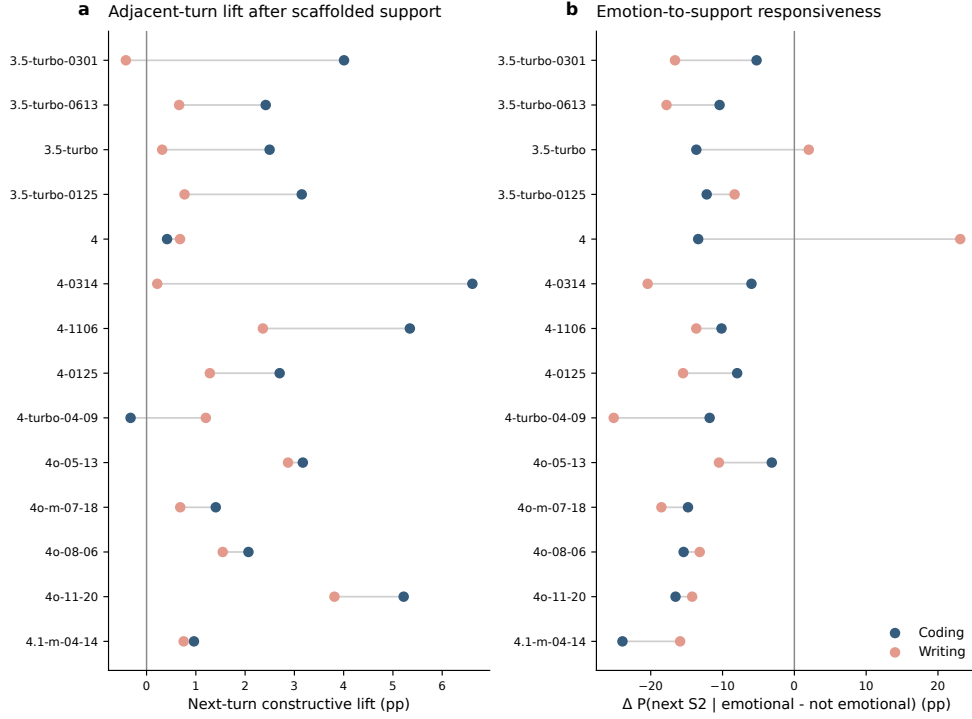


Fig. D6 Local temporal signals vary across WildChat model snapshots. The figure reports adjacent-turn constructive lifts and emotion-to-support responsiveness by model snapshot. Adjacent-turn constructive lifts are often positive, especially in coding-oriented conversations, but the magnitude varies across snapshots; emotion-to-support responsiveness is more heterogeneous and is treated as exploratory.

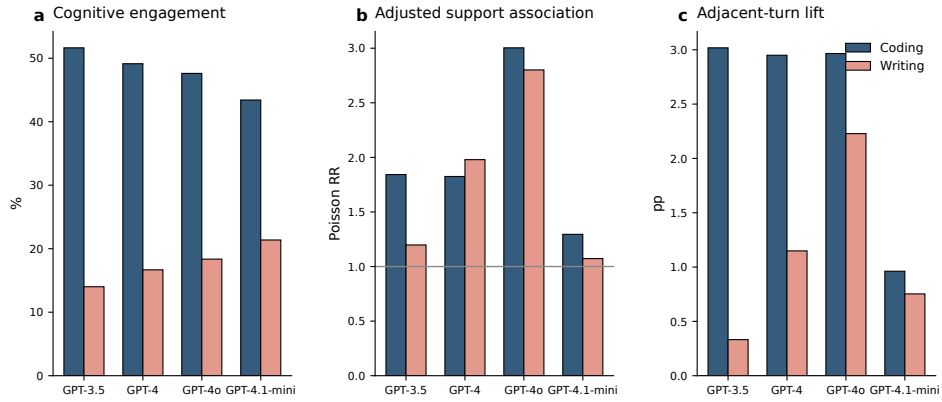


Fig. D7 Coarse model-family aggregation preserves the main directional patterns. WildChat model snapshots are grouped into coarse model families to summarize cognitive engagement prevalence, adjusted support associations and adjacent-turn constructive lift. Coding remains higher than writing across families, and the support-association and local-lift patterns remain directionally consistent on average.